16 Lecture 16

The most important tools for proving theorems concerning almost sure convergence are two simple results: the two Borel–Cantelli lemmas, which we will study in this lecture. To motivate their statements let's think of the following example:

Suppose we repeatedly toss a fair coin, independently, obtaining a sequence of H and T. Consider the events $A_i = \{n\text{-th toss is H}\}$, $i \in \mathbb{N}$. We clearly have $\mathbb{P}(A_i) = 1/2$.

Questions: Will we get infinitely many H? Will we ever get 100 consecutive H?
What if the coin is not fair? What if the probability of getting H becomes smaller and smaller ($\mathbb{P}(A_i) \xrightarrow{i\to\infty} 0$)?

These are some questions that we will be able to answer.

16.1 Borel-Cantelli

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $(A_n)_{n\in\mathbb{N}}$ a sequence of events. We consider the events

$$\limsup_{n\geq 1} A_n = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k = \{\omega \in \Omega : \omega \text{ is in infinitely many } A_n\}$$

and

$$\liminf_{n\geq 1} A_n = \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k = \{\omega \in \Omega : \omega \text{ is eventually in all } A_n\}.$$

We start by showing the 1st Borel-Cantelli lemma, which we indirectly used in Theorem 14.4.

Proposition 16.1 (1st Borel-Cantelli lemma)**.** *Suppose that* $\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty$. *Then*

$$\mathbb{P}(\limsup_{n\geq 1} A_n) = 0.$$

Proof. Let $X = \sum_{n=1}^{\infty} \mathbf{1}_{A_n}$. By Tonelli's theorem,

$$\mathbb{E}[X] = \sum_{n=1}^{\infty} \mathbb{E}[\mathbf{1}_{A_n}] = \sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty,$$

therefore $X < \infty$ almost surely. Since

$$\omega \in \limsup_{n\geq 1} A_n \Leftrightarrow X(\omega) = \infty,$$

we get $\mathbb{P}(\limsup_{n\geq 1} A_n) = 0$. □

Example 2
We consider an urn which at time 0 is empty. At time $n \in \mathbb{N}$ we add to the urn a ball labeled with the number n and we randomly select one of the n balls in the urn (which then contains the balls $1, 2, \ldots, n$). We record the number of the selected ball and return it to the urn. For example, one possible sequence of extracted numbers may begin as follows:

$$1, 2, 2, 1, 4, 3, 6, 2, 7, 9, 8, \ldots$$

We call a *triple* any set of three consecutive positions in the sequence that are occupied by the same number.

Question: Do you expect that in this sequence the number of triples is infinite or finite?

Only For Math 525 Winter 2026
No Permission For Uploading
Has Been Given

Let A_n be the event that the set $\{n, n+1, n+2\}$ forms a triple, that is, the same number is selected at times $n, n+1, n+2$. Then

$$\mathbb{P}(A_n) = \frac{1}{(n+1)(n+2)}.$$

Indeed, this is the probability that the same ball is selected at three consecutive times, hence

$$\mathbb{P}(A_n) = \sum_{k=1}^{n} \frac{1}{n} \cdot \frac{1}{n+1} \cdot \frac{1}{n+2} = \frac{n}{n(n+1)(n+2)} = \frac{1}{(n+1)(n+2)}.$$

Moreover, $\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty$ (**Why?**). The first Borel–Cantelli lemma implies that, with probability 1, only finitely many of the events A_n occur.

Example 3 (inverse of 1st Borel-Cantelli?)
The inverse of the 1st Borel-Cantelli lemma is not true, that is the equality $\mathbb{P}(\limsup_{n\geq 1} A_n) = 0$ does not imply $\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty$. Here is an example.

Let λ be the Lebesgue measure on $\Omega := (0,1)$ (so that λ is a probability measure) and let $A_n = (0, 1/n)$. It is true that $\lambda(A_n) = \frac{1}{n}$, so that the series $\sum_{n=1}^{\infty} \lambda(A_n) = \infty$, while $\limsup_n A_n = \varnothing$. No number belongs to infinitely many of the sets A_n.

The 2nd Borel-Cantelli lemma is related to the case where $\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty$ and the A_n are independent.

Proposition 16.2 (2nd Borel-Cantelli lemma)**.** *Suppose that $(A_n)_{n\in\mathbb{N}}$ is a sequece of independent events such that $\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty$. Then*

$$\mathbb{P}(\limsup_{n\geq 1} A_n) = 1.$$

Proof. We will show that $\mathbb{P}((\limsup_{n\geq 1} A_n)^c) = 1$. We have

$$\mathbb{P}((\limsup_{n\geq 1} A_n)^c) = \mathbb{P}\left(\bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k^c\right) = \lim_{n\to\infty} \mathbb{P}\left(\bigcap_{k=n}^{\infty} A_k^c\right),$$

since the sequence $(B_n)_{n\in\mathbb{N}}$ with $B_n = \cap_{k=n}^{\infty} A_k^c$ is increasing ($B_n \subseteq B_{n+1}$). For this, recall Proposition 3.7.

We now compute due to the independence and the inequality $1 - x \leq e^{-x}$

$$\begin{aligned}\mathbb{P}\left(\bigcap_{k=n}^{\infty} A_k^c\right) &= \lim_{m\to\infty} \mathbb{P}\left(\bigcap_{k=n}^{m} A_k^c\right) = \lim_{m\to\infty} \prod_{k=n}^{m} \mathbb{P}(A_k^c) = \lim_{m\to\infty} \prod_{k=n}^{m} (1 - \mathbb{P}(A_n)) \\ &\leq \lim_{m\to\infty} \prod_{k=n}^{m} e^{-\mathbb{P}(A_k)} = \lim_{m\to\infty} e^{-\sum_{k=n}^{m} \mathbb{P}(A_k)} = e^{-\sum_{k=n}^{\infty} \mathbb{P}(A_k)} = e^{-\infty} = 0\end{aligned}$$

□

Remark 16.3. By Example 3 we see that the independence assumption cannot be dropped in order for the result to hold.

Solution, Example 1. In Example 1, we have that $(A_n)_{n\in\mathbb{N}}$ is an independent sequence with $\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \sum_{n=1}^{\infty} 1/2 = \infty$, hence by the 2nd Borel-Cantelli lemma we have $\mathbb{P}(\limsup_{n\geq 1} A_n) = 1$.

To answer the second question, consider the events

$$B_i = \bigcap_{100(i-1)+1}^{100i} A_i, \quad i \in \mathbb{N}$$

e.g. $B_1 = \{\text{we get heads in tosses 1 to 100}\}, \ldots$. The events B_i are independent and

$$\sum_{i=1}^{\infty} \mathbb{P}(B_i) = \sum_{i=1}^{\infty} \frac{1}{2^{100}} = \infty.$$

Only For Math 525 Winter 2026
No Permission For Uploading
Has Been Given

By the second Borel-Cantelli lemma, we get $\mathbb{P}(\limsup_{i\geq 1} B_i) = 1$, therefore infinitely many of the B_i will happen and, in particular, one B_i will happen.

For the last two questions, the answer truly depends on whether $\sum_{n=1}^{\infty} \mathbb{P}(A_n)$ converges or not. $\square$

Example 4 (almost sure convergence)
Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of i.i.d random variables with density $f(x) = x^{-2}\mathbf{1}_{(1,+\infty)}(x)$. We will show that

$$\limsup_{n\to\infty} \frac{X_n}{n} = \infty, \quad \text{with probability 1.}$$

This equality means that if we repeat this random experiment infinitely many times, then $X_n \geq Mn$ for any $M > 0$ infinitely many times. We will use this last idea as the basis for our proof.

It suffices to show that for any $M > 0$ the event

$$C_M = \left\{\omega \in \Omega : \limsup_{n\to\infty} \frac{X_n(\omega)}{n} \geq M\right\} \quad \text{satisfies} \quad \mathbb{P}(C_M) = 1.$$

Indeed, this gives

$$\mathbb{P}\left(\{\omega \in \Omega : \limsup_{n\to\infty} \frac{X_n(\omega)}{n} = \infty\}\right) = \mathbb{P}\left(\bigcap_{M=1}^{\infty} C_M\right) = 1,$$

because the probability of the intersection of events that have probability 1 has probability 1 (**why?**).
Consider the events $A_n := \{X_n \geq nM\}$, $n \in \mathbb{N}$. Since the random variables X_n are independent, the events A_n are independent and we compute

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \sum_{n=1}^{\infty} \mathbb{P}(X_n \geq nM) = \sum_{n=1}^{\infty} \mathbb{P}(X_1 \geq nM) = \sum_{n=1}^{\infty} \int_{nM}^{+\infty} x^{-2} dx = \sum_{n=1}^{\infty} \frac{1}{nM} = \infty.$$

Therefore, by the 2nd Borel-Cantelli lemma, infinitely many of the A_n happen with probability 1, and hence $X_n/n \geq M$ for infinitely many n with probability 1. The last implies

$$\limsup_{n\to\infty} \frac{X_n}{n} \geq M.$$

Example 5 (almost sure convergence)
Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of i.i.d random variables with $X_1 \sim \text{Exp}(1)$. We will show that

$$\limsup_{n\to\infty} \frac{X_n}{\log n} = 1, \quad \text{with probability 1.}$$

By the definition of lim sup of a sequence this means that for any $\varepsilon > 0$, $X_n/\log n > 1 + \varepsilon$ for finitely many n and $X_n/\log n > 1 - \varepsilon$ for infinitely many n.

For any $n \in \mathbb{N}$ and $r > 0$, we define $A_n^r = \{X_n \geq r \log n\}$. The events A_n^r are independent and satisfy

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n^r) = \sum_{n=1}^{\infty} \mathbb{P}(X_1 \geq r \log n) = \sum_{n=1}^{\infty} e^{-r\log n} = \sum_{n=1}^{\infty} \frac{1}{n^r}.$$

We see that if $r > 1$, then $\sum_{n=1}^{\infty} \mathbb{P}(A_n^r) < \infty$, therefore the 1st Borel-Cantelli lemma implies $\mathbb{P}(\limsup_{n\geq 1} A_n^r) = 0$. Consequently, finitely many of the A_n^r will happen (with probability 1), thus

$$\frac{X_n}{\log n} < r \text{ eventually} \Rightarrow \limsup_{n\to\infty} \frac{X_n}{\log n} \leq r \overset{r\to 1^+}{\Longrightarrow} \limsup_{n\to\infty} \frac{X_n}{\log n} \leq 1 \text{ with probability 1.}$$

Only For Math 525 Winter 2026
No Permission For Uploading
Has Been Given

On the other hand, if $r = 1$, then $\sum_{n=1}^{\infty} \mathbb{P}(A_n^r) = \infty$, hence by the 2nd Borel-Cantelli lemma, infinitely many of the A_n^1 happen. This implies

$$\frac{X_n}{\log n} \geq 1 \text{ for infinitely many } n \implies \limsup_{n\to\infty} \frac{X_n}{\log n} \geq 1 \text{ with probability 1.}$$

Example 6 (Infinite monkey theorem)
Consider an infinite sequence of independent random variables $X_1, X_2, X_3, \ldots$ taking values in a finite alphabet $\mathcal{A}$ (for example the 26 letters, a space and punctuation symbols), each uniformly distributed. Interpret this as a monkey randomly typing characters forever.

Let $w = (w_1, \ldots, w_m)$ be a fixed finite word, for example Shakespeare's Romeo and Juliet. For each $n \geq 1$, define the event

$$A_n = \{X_n = w_1, X_{n+1} = w_2, \ldots, X_{n+m-1} = w_m\},$$

that is, the word w starts at position n. Then

$$\mathbb{P}(A_n) = |\mathcal{A}|^{-m} > 0,$$

and since the A_n are independent for positions spaced at least m apart, we have

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty.$$

By the 2nd Borel–Cantelli lemma, $\mathbb{P}(A_n \text{ occurs infinitely often}) = 1$. Therefore, with probability 1, the monkey will type the word w infinitely many times. Since w was arbitrary, every finite text, including the works of Shakespeare, appears infinitely often almost surely.

Only For Math 525 Winter 2026
No Permission For Uploading
Has Been Given